2000	Established
2001	STATS acquired an approximately 52% equity stake in Winstek.
2005	Approved by the Securities and Futures Bureau of the Financial Supervisory Commission for listing on the OTC market (stock code: 3265) Established a bump production line at a TSMC facility, with a capacity of 10,000 wafers per month
2007	Renamed “StatsChipPac Taiwan Semiconductor Corp.”
2008	Began providing 12-inch electroplated solder bump services
2011	Began providing wafer-level chip-scale packaging (WLCSP) services Relocated the 12-inch bump production line from TSMC Successfully introduced copper pillar bump technology for fine-pitch products and commenced mass production
2015	Following the JCET–STATS merger and acquisition, Temasek became the major shareholder Officially renamed “Winstek Semiconductor Co., Ltd.” on September 30
2017	Sigurd acquired a 52% controlling interest in Winstek (previously held by Bloomberia/Temasek Bloomberg)
2018	Established an 8-inch WLCSP production line, RA/FA laboratories, and an SMT production line
2020	Established a flip-chip production line
2021	Developed advanced semiconductor packaging and testing technologies for silicon photonics
2022	Advanced ESG-driven sustainable operations through ongoing engagement in environmental initiatives and community service activities
2023	Winstek Enterprise Co., Ltd. passed the audit on April 28 and received RBA certification with a Silver rating in May
2024	Published its inaugural Sustainability Report, approved by the Board of Directors and issued concurrently in Chinese and English Winstek Semiconductor Co., Ltd. passed the audit on December 20 and received RBA certification with a Silver rating in January 2025

PBL Phase 2 Research Task

Winstek Business Model: How Can a Semiconductor Company Succeed in Taiwan's AI Era?

Your Main Project Question

Your project is no longer only about the history of Winstek.

Your main research question is:

What is Winstek's business model, how did the company develop its business successfully, and what can we learn from Winstek about building a successful semiconductor company in Taiwan during the AI era?

You already have a good company timeline. **Do not spend most of your next research time adding more dates.** Now use the history you collected to understand **why the company made those changes and how those changes helped its business grow.**

Winstek is mainly a **semiconductor packaging and testing company**, rather than an AI company itself. However, its official 2024 annual report says it has been increasing wafer bumping and wafer-testing business for **5 nm, 4 nm and 3 nm AI/HPC processors**, which gives you a very good connection between Winstek's traditional semiconductor business and the current AI market. (Winstek)

Part 1 — Understand Exactly What Winstek Sells

First answer:

How does Winstek actually make money?

Research each of Winstek's major services and explain them in simple English.

Winstek service	What does Winstek actually do?	Why does a customer need it?	How does this create value?
Semiconductor testing			
WLCSP			
Wafer bumping			
Flip Chip			
Reliability testing			
Failure Analysis			
Other advanced packaging/testing			

Do not simply copy technical definitions. Imagine that you are explaining the company to someone who knows nothing about semiconductors.

For example:

Testing: Winstek checks whether chips work correctly before customers use them in products. This helps customers avoid using defective chips.

Winstek's official site lists WLCSP and testing among its core services, so use the company's own information as an important source.

Part 2 — Find Out Where the Money Comes From

Now investigate Winstek's **revenue/business mix**.

For 2023, 2024 and, if available, newer years, find:

Year	Testing	WLCSP/Packaging	Other services	Total revenue
2023				
2024				
2025/Latest available				

Then answer:

Which part of Winstek's business is becoming larger?

Which part is becoming smaller?

Why might this change be happening?

Does the growth of AI/HPC appear to be connected to these changes?

Be careful: if you think AI caused a change, **find evidence before saying that it caused it**.

The 2024 annual report is especially useful because it provides Winstek's financial and market information and discusses AI/HPC demand. (Winstek)

Part 3 — Who Are Winstek's Customers?

Find out: **Who needs Winstek?**

You may not always be able to find the names of individual customers because semiconductor companies often keep customer information confidential.

Therefore, also research **types of customers**, such as:

Chip-design companies → wafer manufacturers → packaging/testing companies → electronics manufacturers → final customers

Determine exactly **where Winstek fits into this chain**.

Create your own semiconductor supply-chain diagram:

Chip design

→ Wafer manufacturing

→ Winstek: bumping / packaging / testing

→ Electronics manufacturer

→ Final product

Then investigate which products could eventually use these chips:

📱 Smartphones

🖥️ AI servers

🚗 Vehicles

📡 Communication equipment

⌚ Wearable devices

☁️ Data centers

🤖 AI systems

Your question is: **Why would another semiconductor company pay Winstek instead of doing all of the packaging and testing itself?**

Part 4 — What Is Winstek's Value Proposition?

This is one of the most important parts of a **business model**.

A value proposition means:

What useful thing does Winstek provide that customers are willing to pay for?

Research whether Winstek creates value through areas such as:

advanced technology + quality + reliability + speed + specialized equipment + experienced engineers + smaller packaging + advanced testing + customer support

Do not assume all of these are true. Find evidence.

Then complete this sentence:

“A semiconductor company chooses Winstek because _____.”

Try to produce a clear answer supported by evidence.

Part 5 — Why Does Technology Matter to the Business?

Use your original timeline here.

Do not only write:

2011 — WLCSP introduced.

Instead ask:

Why was introducing WLCSP important for Winstek's business?

Do this for several major changes:

Technology/change	What changed?	Why was it important for customers?	How could it help Winstek's business?
Wafer bumping			
12-inch production			
WLCSP			
Copper pillar			
Flip Chip			
Reliability/Failure Analysis			
Silicon Photonics			
AI/HPC testing			

Your timeline will therefore change from a **history timeline** into a **business-development timeline**.

Part 6 — Why Are AI and HPC Important to Winstek?

This should become an important part of your project.

Research:

What is an AI chip?

What is HPC?

Why do AI/HPC chips require advanced semiconductor technology?

Why do increasingly advanced chips also create opportunities for packaging and testing companies?

What types of testing might powerful AI chips require?

What are 5 nm, 4 nm and 3 nm technologies?

Why is Winstek increasing its business related to these advanced processors?

Winstek's 2024 report specifically says the company increased its proportion of wafer bumping and testing orders for **5 nm, 4 nm and 3 nm AI/HPC processors**, and also mentions growth areas including Wi-Fi 7, GaN, HPC and automotive-chip certification. (Winstek)

Your goal is to explain the connection:

AI growth → demand for advanced chips → more complex semiconductor manufacturing → need for advanced packaging/testing → business opportunity for companies such as Winstek

But verify each part of this relationship with good sources.

Part 7 — Research Silicon Photonics

Investigate why Winstek has been developing semiconductor packaging/testing capability related to **silicon photonics**.

Answer:

What is silicon photonics?

What problem does it try to solve?

Why are very fast data connections important for AI and data centers?

Why might semiconductor testing companies need new testing technology for optical devices?

Could silicon photonics become an important future business for Winstek?

Separate clearly between:

What Winstek is already doing

and

what you think could become a future opportunity.

Part 8 — What Does Winstek Need in Order to Operate?

A successful semiconductor company cannot operate with technology alone.

Research the important **resources** Winstek needs.

Think about:

Resource	Why is it important?
Engineers and technicians	
Semiconductor equipment	
Factory/cleanroom	
Electricity	
Water	
Chemicals/materials	
Testing equipment	
R&D	
Software/AI	
Suppliers	
Money/investment	

Then answer:

Which resources are the hardest or most expensive for a new semiconductor company to obtain?

This is important because your final project is trying to understand **how to create a successful company**, not simply how semiconductor technology works.

Part 9 — Where Does Winstek Spend Its Money?

Investigate the company's important costs.

Look particularly for:

Capital expenditure (CapEx) — money spent on new machines, equipment, facilities and production capacity.

Also research:

Employee costs, electricity, water, materials, maintenance, R&D, depreciation and environmental management.

Try to answer:

Why must a semiconductor company continue buying new equipment?

What happens if it buys too much equipment but customer demand decreases?

What does depreciation mean for a semiconductor business?

Why is investment risky but necessary?

This will help you understand that a successful semiconductor company needs both **good technology and good financial decisions**.

Part 10 — Taiwan: Why Build the Company Here?

Now move beyond Winstek.

Research:

Why is Taiwan an important location for a semiconductor company?

Investigate factors such as:

semiconductor supply chain, skilled workers, universities, engineering talent, nearby suppliers, customers, transportation, Hsinchu Science Park and the concentration of semiconductor companies.

Then ask a more specific local question:

Why might Hsinchu County be useful for a semiconductor company such as Winstek?

Do not just say “because Taiwan makes semiconductors.” Explain **what advantages the ecosystem provides to a business.**

Part 11 — Competition

Find several companies that provide similar semiconductor packaging/testing services.

Do **not** simply decide who is “better.”

Instead compare factual features such as:

Company	Main services	Technology areas	Main markets	Size/revenue if available	AI/HPC-related work
Winstek					
Competitor 1					
Competitor 2					

Then answer:

How can a smaller company compete when much larger semiconductor companies already exist?

This is an important business question.

Part 12 — Sustainability as Part of the Business

Semiconductor factories use significant resources, so investigate whether sustainability has become part of Winstek's business strategy.

Look at:

energy use, carbon emissions, water, waste recycling, supplier requirements, worker safety and environmental management.

Winstek reports initiatives including energy-management certification, greenhouse-gas verification, WLCSP carbon-footprint verification and supplier sustainability requirements.

Ask:

Can sustainability reduce costs?

Can environmental requirements affect customers' decisions when choosing suppliers?

Can sustainability therefore become part of a semiconductor company's business model rather than simply charity?

Part 13 — Complete a Winstek Business Model Canvas

After completing your research, create **one final Business Model Canvas**.

Business Model Section	Your Winstek Research
Customer Segments	Who needs Winstek?
Value Proposition	Why do customers choose/pay Winstek?
Key Services	What does it sell?
Key Activities	What work happens inside the company?

Key Resources	What does the company need to operate?
Key Partners	Who does it depend on?
Revenue Streams	Where does its money come from?
Cost Structure	What does it spend money on?
Technology Strategy	What technology keeps it competitive?
Future Opportunities	AI/HPC, silicon photonics, etc.
Risks	What could damage the business?
Sustainability	How does ESG/environmental management affect operations?

For **every important statement**, provide a source.

Your Final Challenge

After studying Winstek, answer:

"If I wanted to build a successful semiconductor company in Taiwan for the AI era, what would I need?"

Do not just give an opinion.

Create your answer from your research.

For example, investigate whether a successful company would need:

1. A real customer problem to solve

2. Specialized semiconductor technology

3. Skilled engineers

4. Strong quality control

5. Expensive but carefully planned equipment investment

6. Relationships with customers and suppliers

7. Continuous R&D

8. Ability to follow new markets such as AI/HPC

9. Financial management

10. Environmental and sustainability management

Your research should determine which factors are actually important.

Exhibition Product — "Build Your Own Semiconductor Company"

For your exhibition, create an **interactive business model**, not only a poster.

Make a large board or 3D model titled:

CAN YOU BUILD A SUCCESSFUL AI-ERA SEMICONDUCTOR COMPANY IN TAIWAN?

Give each visitor **10–20 investment tokens**.

They become the CEO of a new semiconductor company.

They must decide how much to invest in areas such as:

🏭 Factory & equipment

🔬 R&D

👷 Engineers

🧪 Testing & quality control

📦 Advanced packaging

🤖 AI/HPC technology

💡 Silicon photonics

🌱 Sustainability

📈 Marketing/customers

⚡ Energy and infrastructure

But they **cannot afford everything**.

After they make their choices, your exhibition should explain the consequences.

For example:

High technology + no trained workers → Problem

Huge factory + insufficient customers → Financial risk

Many customers + poor testing → Quality problem

Strong R&D + engineers + customers + controlled investment → More sustainable business

The point is **not to claim there is one perfect formula**. The activity should show visitors the trade-offs a semiconductor business faces.

Final Presentation Structure

Your final exhibition story should be:

1. Where Winstek started

↓

2. How its technology changed

↓

3. What Winstek sells today

↓

4. How Winstek makes money

↓

5. Why customers need it

↓

6. How AI/HPC are changing its opportunities

↓

7. What risks and costs the company faces

↓

8. What Winstek teaches us about building a semiconductor business in Taiwan

↓

9. Visitor builds their own semiconductor company

Good Sources to Start With

Use **official sources first**, especially Winstek's **Annual Reports**, **Sustainability Reports**, product/service pages and financial information. Winstek provides its reports and sustainability materials directly on its website.

You may also use Sigurd's official information because Sigurd is Winstek's parent-group company and publishes information about packaging/testing technologies and AI/automation applications. (Sigurd)

Do not rely mainly on YouTube videos, blogs or investment-commentary websites. They can give you ideas, but verify important claims using official reports or other reliable sources.

Your goal

By the end of this project, a visitor should be able to ask you:

“So what actually makes a semiconductor company successful?”

and you should be able to answer using **Winstek as a real Taiwan case study, financial/business evidence, technology evidence, and your own analysis**—not simply information copied from the internet.